

APPLICATION GUIDE

Spray Drying Applications Guide

A material-focused overview of process challenges, powder targets and equipment directions.

Six industries | Common process challenges | Scale selection

<http://oltdryer/>



Equipment selection begins with the feed and the required powder



1

What is the feed?

Solution, suspension, slurry or emulsion

2

What makes it difficult?

Heat, stickiness, viscosity, solvent, corrosion or dust

3

What powder is required?

Moisture, particle size, density, flow and solubility

4

What scale is next?

Research, pilot validation, small batch or production

Pharmaceuticals



CONTROLLED DEVELOPMENT

Spray drying can support formulation development where thermal exposure, containment, repeatability and cleaning require careful review.

Representative feeds

- APIs and excipients
- Formulation solutions
- Biologics and specialty powders

Common challenges

- Heat sensitivity
- Residual moisture
- Containment and cross-contamination

PROCESS DIRECTION

Controlled operating conditions, suitable atomization, collection review and documented cleaning direction.

Food & Beverage



QUALITY AND RECOVERY

Food applications connect aroma, solubility, ingredient stability and powder handling with hygienic equipment requirements.

Representative feeds

- Coffee and tea extracts
- Flavors and natural colors
- Fruit concentrates and proteins

Common challenges

- Aroma retention
- Wall sticking
- Powder flow and solubility

PROCESS DIRECTION

Review formulation support, temperature window, atomization and efficient powder collection.

Nutraceuticals



ACTIVE INGREDIENT PROTECTION

Botanicals, vitamins and functional ingredients often need a balanced approach to active retention, carrier selection and storage stability.

Representative feeds

- Herbal and botanical extracts
- Vitamins and functional blends
- Probiotic or specialty ingredients

Common challenges

- Thermal degradation
- Stickiness and low recovery
- Flowability and stability

PROCESS DIRECTION

Evaluate carriers, thermal tolerance, collection behavior and a suitable validation scale.

Chemicals



COMPATIBILITY AND CONTROL

Chemical feeds require review of composition, corrosion, solvent basis, dust characteristics and the target particle profile.

Representative feeds

- Fine chemicals and additives
- Catalyst or pigment suspensions
- Polymer and specialty solutions

Common challenges

- Corrosion and compatibility
- Solvent or explosion risk
- Particle control and collection

PROCESS DIRECTION

Confirm contact materials, atomization, collection, filtration and project-specific safety requirements.

Agrochemicals



FORMULATION PERFORMANCE

Agrochemical development may use spray drying to convert formulated liquids or suspensions into stable, handleable powders.

Representative feeds

- Pesticide formulations
- Adjuvant systems
- Biological and mineral feeds

Common challenges

- Suspension stability
- Dust and operator exposure
- Redispersion and storage stability

PROCESS DIRECTION

Review feed agitation, atomization, containment, collection and the required powder handling properties.

Ceramics & Advanced Materials



PARTICLE ENGINEERING

Advanced-material feeds connect dispersion quality and atomization with granule size, morphology, bulk density and downstream processing.

Representative feeds

- Ceramic and mineral slurries
- Metal oxides and precursors
- Battery and functional materials

Common challenges

- Sedimentation and abrasion
- Particle distribution
- Moisture and contamination control

PROCESS DIRECTION

Evaluate feed preparation, wear-resistant construction, atomization and product collection around the downstream process.

Material behavior defines the engineering questions



Heat sensitivity

Control thermal exposure and outlet conditions



Stickiness

Review formulation, deposition and collection



Organic solvents

Evaluate closed-loop recovery, inerting and safety



Particle size

Connect atomization with drying history and collection

A material can present more than one challenge. Testing helps establish which constraints control the process direction.

Choose the next scale according to the decision you need to make



Laboratory

1.5-3.0 L/h

Feasibility, formulation screening and initial powder observation



Pilot

5-30 L

Longer validation runs, optimization and representative data



Industrial

50-100+ L

Production duty, continuous operation and plant integration

Material data turns a broad application into a testable process question



Feed properties	Solids, viscosity, pH, density and stability
Thermal behavior	Temperature tolerance and active protection
Solvent basis	Water, organic solvent or mixed solvent
Powder target	Moisture, size, flow, density and solubility
Capacity	Available sample, evaporation duty and production goal

Share the material, target powder and project stage

Engineer-reviewed direction

Submit the information you have. Unknown values can be clarified before testing or equipment selection.

Material evaluation

Pilot testing

Equipment proposal

Start Your Material Evaluation

<http://olldryer/material-evaluation/>

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